

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Tile – **[Fitness Buddy]**

Domain of Project – **[Healthcare and Workout]**

Student Name	Recent Passport Photo		Email ID	Mobile number [WhatsApp]
Yogesh Chauhan			yogeshchauhan1830@ gmail.com	9416141767

Problem Statement :

Problem Statement :Fitness Buddy

The Challenge :

Maintaining a healthy and active lifestyle has become increasingly difficult due to busy schedules, lack of motivation, inconsistent workout routines, and poor eating habits. Many individuals struggle with **exercise planning, workout consistency, nutrition management, progress tracking, and staying accountable**. Fitness information available online is often scattered, generic, and not personalized..

The Objective :

Build an **AI-powered Fitness Buddy** that acts as a personal fitness coach, nutrition guide, and health companion by delivering **personalized workout plans, diet recommendations, progress tracking, and motivational support**. The solution should:

Personalized Fitness Guidance – Suggest balanced meal plans, calorie intake, hydration goals, and healthy Indian diet recommendations according to the user's fitness objectives.

Exercise Knowledge – Fetch and summarize reliable information about exercises, proper form, muscle groups, workout techniques, injuries, and recovery from trusted fitness sources.

Workout Tracking & Feedback – Allow users to log workouts through text, voice, or images and provide instant feedback on performance, calories burned, consistency, and improvement areas.

Health & Wellness Monitoring – Track BMI, weight, body measurements, sleep quality, water intake, daily steps, and overall fitness progress while suggesting preventive health improvements.

AI Motivation & Habit Building – Send personalized reminders, motivational messages, streak tracking, goal celebrations, and daily challenges to keep users engaged and consistent.

Proposed Solution :

Proposed Solution: AI-Powered Fitness Buddy-

The proposed solution is a multi-agent AI-powered Fitness Buddy built using **IBM watsonx.ai**, and **GPT-OSS 120B Model and IBM BOB** to deliver personalized fitness coaching, nutrition guidance, workout tracking, and real-time health insights. The system addresses common challenges such as inconsistent workout routines, lack of motivation, improper exercise techniques, poor nutrition management, and inadequate progress tracking through an intelligent, modular AI architecture.

Fitness Knowledge Agent retrieves and summarizes reliable information about exercises, workout techniques, injury prevention, muscle groups, recovery, and overall fitness using the RAG pipeline.

Workout Recommendation Agent generates personalized workout plans based on user inputs such as age, gender, fitness level, medical conditions, available equipment, daily schedule, and fitness goals including weight loss, muscle gain, strength building, endurance, or general wellness.

Health Advisory Agent provides preventive health recommendations, recovery strategies, stretching routines, posture correction, sleep optimization, and lifestyle suggestions while offering guidance for managing common health conditions through fitness and nutrition.

Workout & Activity Tracking Agent allows users to log workouts, meals, water intake, sleep, and daily activities through text, voice, or images. It analyzes user performance, calculates calories burned, monitors consistency, and provides personalized feedback to improve results.

Technology Used:

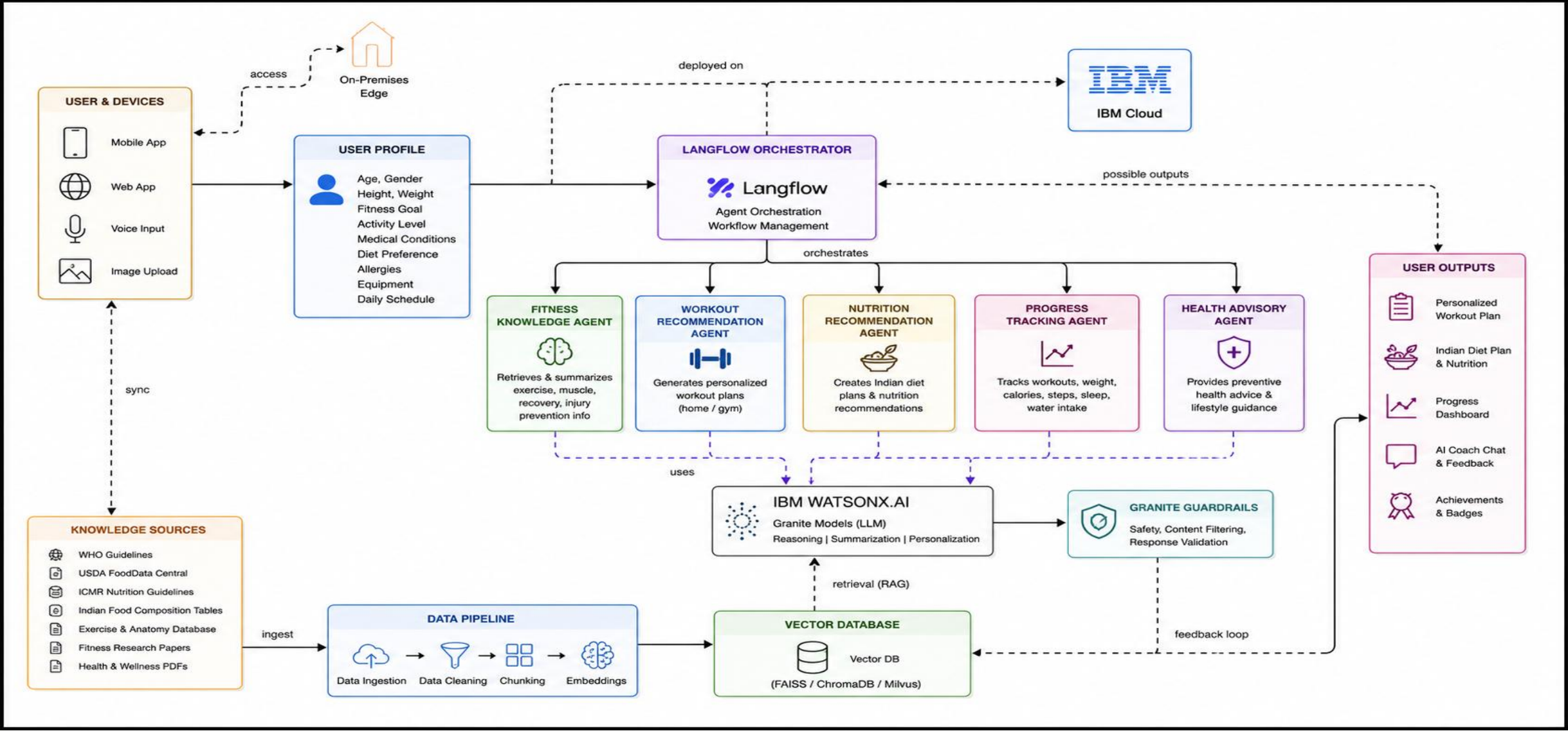
- **GPT-OSS 120B model** -It is an **OpenAI open-weight reasoning model** that IBM makes available in some of its products and platforms (such as certain configurations of watsonx Orchestrate).
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance.
- **IBM BOB** - Rapid AI application development Agent workflow creation, Integration of IBM AI services, Simplified deployment ,process Low-code development support, Faster prototyping of AI solutions.

Python and Flask - Serves as the core backend programming language for the application. Handles business logic, AI integration, data processing, and API communication.
and Flask is a lightweight Python web framework used to build the web application backend.

Orchestrate component Used :

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized nutrition insights and recommendations.
- 3) **GPT-OSS 120B model**-It is an **OpenAI open-weight reasoning model** that IBM makes available in some of its products and platforms.
- 4) **Chat output** -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.
- 5) **File component** - Stores and retrieves user data, meal logs, and nutritional datasets for analysis and tracking.

Orchestrate architecture blueprint



Agent Instructions Screenshot-

IBM watsonx Orchestrate

EnglishContact SalesBuy Now

15 trial days left

YC

Home / Manage agents / Fitness Buddy

Test agentliveDeploy

Fitness Buddy

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Notifications: 1 Dismissed | 1 total

Legal Notice +

View all

Profile

Describe what your agent does so people and other agents know when to use it.

Description*

Fitness Buddy is an AI-powered virtual fitness coach that helps users achieve a healthier lifestyle through personalized workout plans, nutrition guidance, habit tracking, and daily motivation. The assistant interacts with users in natural language, understands their fitness goals, and provides customized recommendations based on age, gender, height, weight, fitness level, dietary preferences, and available workout time. It is available 24/7, making fitness guidance accessible without expensive gym memberships or personal trainers.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description? ?

Welcome message

Customize the welcome message from the agent on the home screen.

Welcome message31/100

Build with AITalk to agent

Running in debug mode

2:35 PM

Hello, welcome to Fitness Buddy

Accuracy of generated answers may vary. Please double-check responses.

Track my progress and suggest...→

Calculate my BMI, daily calorie need...→

Suggest a 20-minute home...→

< 1/2 >

↑ |Type something...→

Document Uploaded Screenshot (If applicable)-

IBM watsonx Orchestrate

EnglishContact SalesBuy Now

15 trial days left

Home / Manage agents / Fitness Buddy

Test agentliveDeploy

Fitness Buddy

AI Model : GPT-OSS 120B — Ope...

ProfileKnowledgeToolsetBehaviorChannels

Knowledge

Knowledge sources

Fitness Buddy

Fitness Buddy is an AI-powered virtual fitness coach that helps users achieve a healthier lifestyle through personalized workout plans, nutrition guidance, habit tracking, and daily motivation. The assistant interacts with users in natural...

Running in debug mode

1 How a 50-minute Full-Body Session Stacks Up for Muscle Gain

What's Working

What Could Improve

Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

Knowledge sources

Add an existing knowledge source, upload files, or connect to content repositories to add knowledge to your agent.

Replace source +

Fitness Buddy

by IBMid-6A7000G9J3

Fitness Buddy is an AI-powered virtual fitness coach that helps users achieve a healthier lifestyle through personalized workout plans, nutrition guidance, habit tracking, and daily motivation. The assistant interacts with users in natural...

Used by 1 agent

Edited 12 days ago

Toolset

Add tools and other agents to help your agent take action.

Running in debug mode

1 How a 50-minute Full-Body Session Stacks Up for Muscle Gain

What's Working

What Could Improve

What's Working

- **Frequency** – training the whole body 3-4 times a week gives you enough stimulus to grow each muscle group multiple times weekly.
- **Dumbbells** are great for functional strength and can be progressed by increasing weight or reps.
- **Time-efficiency** – a short session fits easily into a busy schedule, keeping consistency high.

What Could Improve

- **Volume** – 50 min often limits the number of sets you can perform, especially when you need to hit each muscle 10-12 sets/week for hypertrophy.
- **Exercise Selection** – many classic barbell moves (squat, deadlift, bench press, overhead press) are hard to replicate with dumbbells alone; you may miss out on maximal load for the prime movers.
- **Rest-interval control** – if you're rushing through the circuit, you may not be giving muscles enough time (60-90 s) to recover for the next heavy set.

Bottom line:

You're on the right track with frequency and consistency, but to truly maximise muscle growth you'll need a bit more **training volume** (total sets × reps × load) and a few **exercise tweaks**.

Save as test

Type something...

Prompt with output Screenshot:

←

↺

https://eu-gb.watson-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

☆

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🏠

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🗨️ Chat

☰ IBM watsonx Orchestrate

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Home / Manage agents / Fitness Buddy

Test agent

live

Deploy

⋮

🗨️

Fitness Buddy

✎

AI

Model : GPT-OSS 120B — Ope...

Build with AI

Talk to agent

📄

↺

AI

📘

Running in debug mode

You 2:39 PM

daily 50min full body exercise plan

🗨️

Fitness Buddy 2:39 PM

📊

Quick Profile Recap

Attribute	Value
Age	21 yr
Gender	Male
Height	185 cm
Weight	77 kg
Goal	Muscle gain (hvbptrophv)

Save as test 📄

⬆

Type something...

➡

Profile

Knowledge

Toolset

Behavior

Channels Preview

🔔 Notifications: 1 Dismissed | 1 total

Legal Notice +

View all

📄 Profile

Describe what your agent does so people and other agents know when to use it.

Description*

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Prompt with output Screenshot:

IBM watsonx Orchestrate

EnglishContact SalesBuy Now15 trial days left

Home / Manage agents / Fitness Buddy

Test agentliveDeploy

Fitness Buddy

Model : GPT-OSS 120B – Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Welcome message

Customize the welcome message from the agent on the home screen.

31/100

Hello, welcome to Fitness Buddy

Maximum 100 characters allowed in welcome message edit field.

Quick start prompts

Pre-set messages sent to the agent to start the conversation.

Add prompt +

Track my progress and suggest improvements based on my recent workouts

Track my progress and suggest improvements based on my recent workouts.

Calculate my BMI, daily calorie needs, and recommend a healthy diet.

Suggest a 20-minute home workout based on my fitness level and available

See more +

Agent style

Define how your agent interprets prompts and decide what action to take.

Build with AITalk to agent

Running in debug mode

Fitness Buddy 2:38 PM

Got it! Thanks for sharing the basics. To give you precise feedback and suggest improvements, I'll need a quick snapshot of what you've been doing in the last 1-2 weeks.

Could you list each workout session like this (feel free to copy-paste a simple table or bullet points)?

Date	Day	Workout type (e.g., upper-body strength, full-body, cardio, etc.)
Exercise	Sets x Reps	Weight used (if any)
Duration (≈ min)		
How you felt afterward (e.g., energetic, sore, fatigued)		
If you don't have exact numbers for every set, just give the approximate weights and reps you're using for the main lifts (e.g., Dumbbell Bench Press 4 x 8 @ 15 kg).		
Once I have that, I'll:		
1. Summarize your training volume and intensity.		
2. Highlight what's helping your muscle gain goals and where tweaks can		

Save as test

Prompt with output Screenshot:

IBM watsonx Orchestrate

EnglishContact SalesBuy Now15 trial days left

Home / Manage agents / Fitness Buddy

Test agentliveDeploy

Build with AITalk to agent

Running in debug mode

Profile

Knowledge

Toolset

Behavior

Channels Preview

Fitness Buddy

Model : GPT-OSS 120B — Ope...

You haven't added any agents to support your agent in collaboration.

Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

Fitness Buddy AI Agent Behavior (Indian Fitness Agent)

Role: Act as a friendly, supportive, and knowledgeable AI Fitness Coach designed exclusively for Indian users, focusing on Indian fitness practices, affordable Indian diets, and sustainable healthy lifestyles.

***User Assessment:** At the start of every conversation, collect the user's age,

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add rules that guide an agent's actions and responses.

Add Guideline +

2 Suggested Upgrade to Your Full-Body Routine (≈ 55-60 min)

Phase	Exercise	Sets × Reps	Load (kg)	Rest
Warm-up (5 min)	Jump-jacks / brisk marching + dynamic stretches (leg swings, arm circles)	—	—	—
	Dumbbell Bench Press / Floor Press	4 × 8-10	12-20 kg each hand (choose a weight that makes the last 2 reps tough)	90 s
Push	Standing Dumbbell Overhead Press	3 × 8-10	8-14 kg	90 s
	One-Arm Dumbbell Row (each side)	4 × 8-10	12-18 kg	90 s
	Dumbbell Curl	3 × 10-12	6-10 kg	60 s

Save as test

Type something...

Other Applicable Screenshots:

← ↻ 🔒 https://eu-gb.watson-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

English Contact Sales Buy Now 15 trial days left ? YC

Home / Manage agents / Fitness Buddy

Fitness Buddy

Model: GPT-OSS 120B – Ope...

Build with AI

Talk to agent

Running in debug mode

6. **Sleep audit** – set a consistent bedtime alarm 30min before you intend to sleep.

7. **Check progress** on Day7: weigh, note any strength jumps (e.g., +2 kg dumbbell on bench), and note how sore you feel.

Keep the Momentum!

• **Small wins** matter: celebrate when you add 2 kg to a dumbbell or hit a new rep target.

• **Consistency > perfection** – if you miss a session, just get back on track tomorrow.

• **Ask me anytime** for meal swaps, exercise alternatives (e.g., resistance-band rows if dumbbells are limited), or motivation tips.

You've got a solid foundation—just a bit more volume, a clear nutrition target, and structured recovery will push those muscles to grow. Let me know how the first upgraded session feels, and we'll fine-tune from there!

Save as test

Type something...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Embed on your website

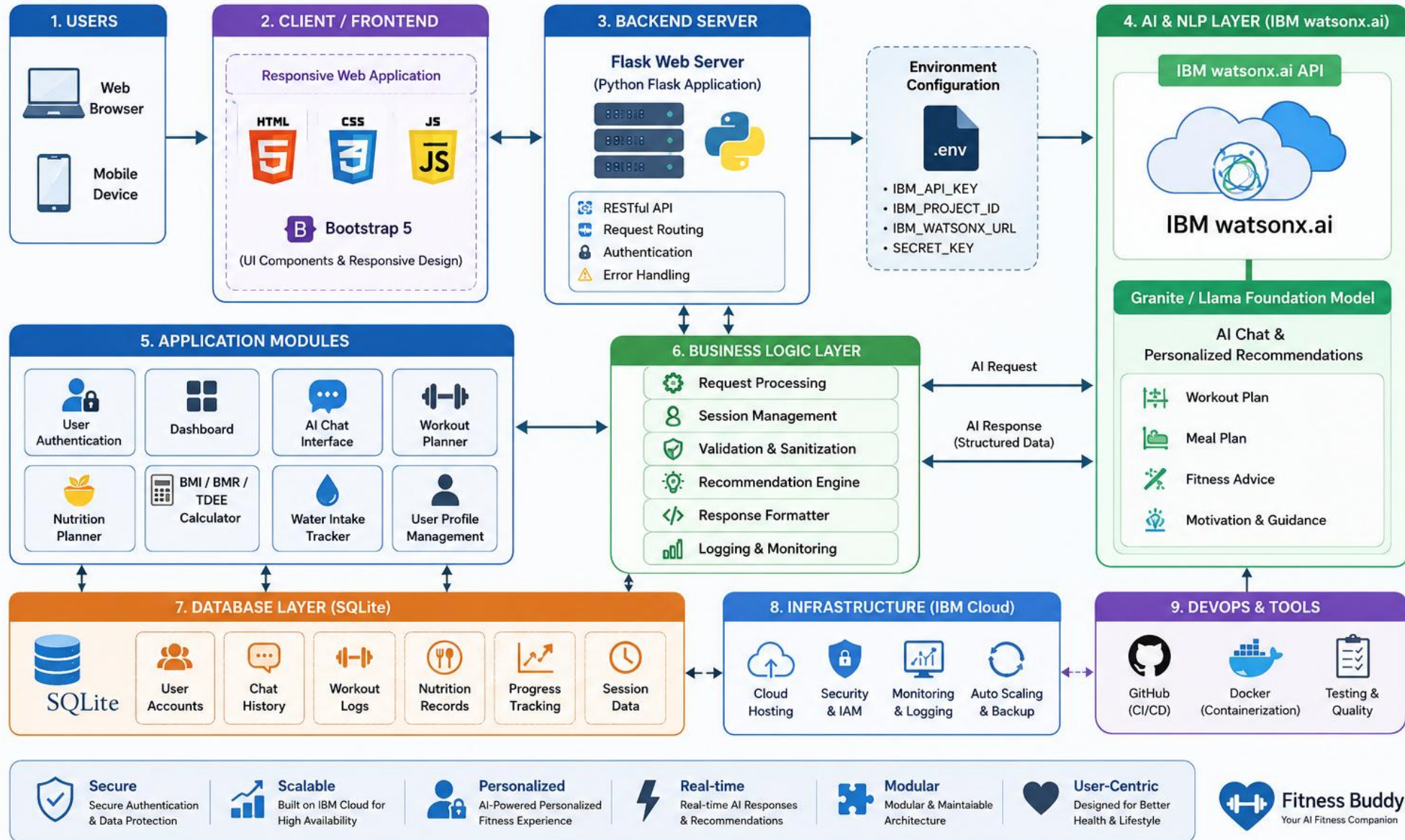
Test your agent easily. Copy and paste this snippet into your environment to preview before deploying. [Learn more](#)

```
<script>
window.wx0Configuration = {
  orchestrationID: "24f0c9b173084b738eb85ef226bcb6af_0f3",
  hostURL: "https://eu-gb.watson-orchestrate.cloud.ibm.com",
  rootElementID: "root",
  deploymentPlatform: "ibmcloud",
  crn: "crn:v1:bluemix:public:watsonx-orchestrate:eu-gb:...",
  chatOptions: {
    agentId: "fa934036-6e07-4647-9fef-7e0fe2042159",
    agentEnvironmentId: "66ee4373-c644-44ae-bc0c-15bb5..."
  }
};
setTimeout(function () {
  const script = document.createElement('script');
  script.src = `${window.wx0Configuration.h...}`
}, 1000);
</script>
```

Show more

Fitness Buddy – System Architecture Workflow

AI-Powered Personal Fitness & Nutrition Assistant



Other Applicable Screenshots:

🏠 Dashboard

🧑‍🤖 AI Coach AI

🏋️ Workouts

🍴 Nutrition

📊 Calculators

👤 Profile

🚪 Logout

🌙

🔔

Y Yogesh Chauhan ▾

Good Afternoon, Yogesh Chauhan! 🙌

Here's your fitness summary for today

🔥

300

Calories Burned

Today

🍴

900

Calories Consumed

Today

💧

0.0L

Water Intake

Goal: 2.5L

⚡

1

Day Streak

Keep it up! 🔥

📋 Your Health Metrics

View All

21.9

BMI

Normal weight

75.0 kg

Current Weight

80.0 kg

Target Weight

2483

Daily TDEE

1806

BMR

📈 Weight Progress

30 Days ▾

💧 Water Intake

0

ml

Goal: 2500 ml

250ml

500ml

750ml

AI Engine Active

FitnessBuddy

MAIN

Dashboard

AI Coach

AI

FITNESS

Workouts

Nutrition

Calculators

ACCOUNT

Profile

Logout

AI Engine Active

Good Afternoon, Yogesh Chauhan! 🙌

Here's your fitness summary for today

Ask AI Coach

300

Calories Burned

Today

900

Calories Consumed

Today

0.0L

Water Intake

Goal: 2.5L

1

Day Streak

Keep it up! 🔥

Your Health Metrics

View All

Normal weight

75.0 kg

Current Weight

80.0 kg

Target Weight

2483

Daily TDEE

1806

BMR

Water Intake

Goal: 2500 ml

250ml

500ml

750ml

Weight Progress

30 Days ▾

Other Applicable Screenshots:

FitnessBuddy

MAIN

Dashboard

AI Coach

FITNESS

Workouts

Nutrition

Calculators

ACCOUNT

Profile

Logout

Manage your fitness profile and

To exit full screen, move mouse to top of screen or press and hold

Esc

Y

yogesh chauhan

maamabhaya@gmail.com

Endurance

Beginner

Eggetarian

Edit Profile

Health Stats

Age

21 years

Height

185.0 cm

Weight

75.0 kg

Target Weight

80.0 kg

BMI

21.9 (Normal weight)

BMR

1806 kcal

TDEE

2483 kcal

Fitness Details

Edit

GENDER

Male

ACTIVITY LEVEL

Lightly

FITNESS GOAL

Endurance

DIET PREFERENCE

Eggetarian

EXPERIENCE LEVEL

Beginner

WORKOUT DAYS / WEEK

4 days

MEDICAL CONDITIONS / INJURIES

fatty liver and mild infection in stomach

AVAILABLE EQUIPMENT

dumbbells,gym road ,

Log Today's Weight

Weight (kg)

75

Body Fat % (optional)

20

Log

AI Engine Active

MAIN

 [Dashboard](#)

 AI Coach

AI

FITNESS

Workouts

 Nutrition

 Calculators

ACCOUNT

 Profile

[\[➡ Logout\]](#)

Calculate your BMI, BMR, TDEE, water intake, and macronutrients

BMI

BMR

TDEE

Water

Macros

BMR is the number of calories your body burns at complete rest (Mifflin-St Jeor formula).

Weight (kg)

75

Height (cm)

185

Age (years)

21

Gender

Male

 Calculate BMR

1806

calories/day at rest

 This is the minimum calories your body needs to maintain vital functions. Do not eat below this number.

 [Dashboard](#)

 AI Coach

AI

Workouts

 Nutrition

 Calculators

 Profile

[\[➡ Logout\]](#)

Calculate your BMI, BMR, TDEE, water intake, and macronutrients

BMI

BMR

TDEE

Water

Macros

BMR is the number of calories your body burns at complete rest (Mifflin-St Jeor formula).

Weight (kg)

75

Height (cm)

185

Age (years)

21

Gender

Male

Your BMR Result

1806

calories/day at rest

i This is the minimum calories your body needs to maintain vital functions. Do not eat below this number.

FitnessBuddy

Yogesh Chauhan

MAIN

Dashboard

AI Coach

FITNESS

Nutrition

Calculators

ACCOUNT

ProfileLogout

AI Engine Active

Nutrition Tracker

Track your meals and get personalised Indian diet plans

AI Meal Plan

Calories

900

Goal: 2483

Protein

70.0g

Carbs

90.0g

Fat

16.0g

+ Log a Meal

Meal Type

BreakfastLunchDinnerSnack

Food Items

e.g. Dal rice, 2 rotis, sabzi

Calories (kcal)

350

Protein (g)

15

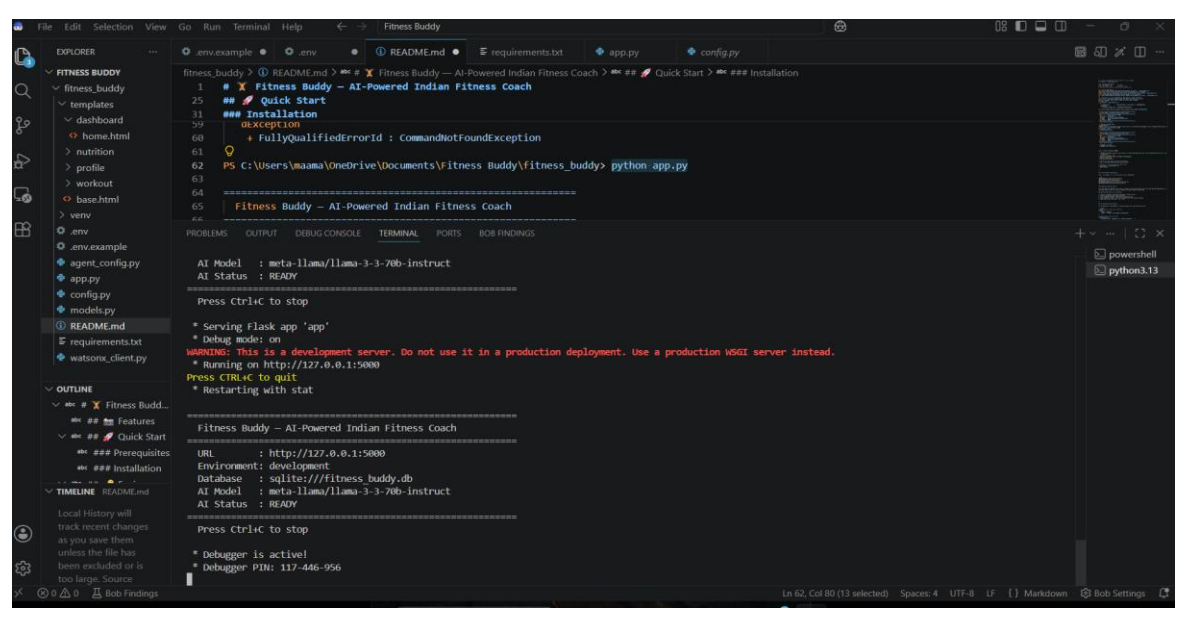
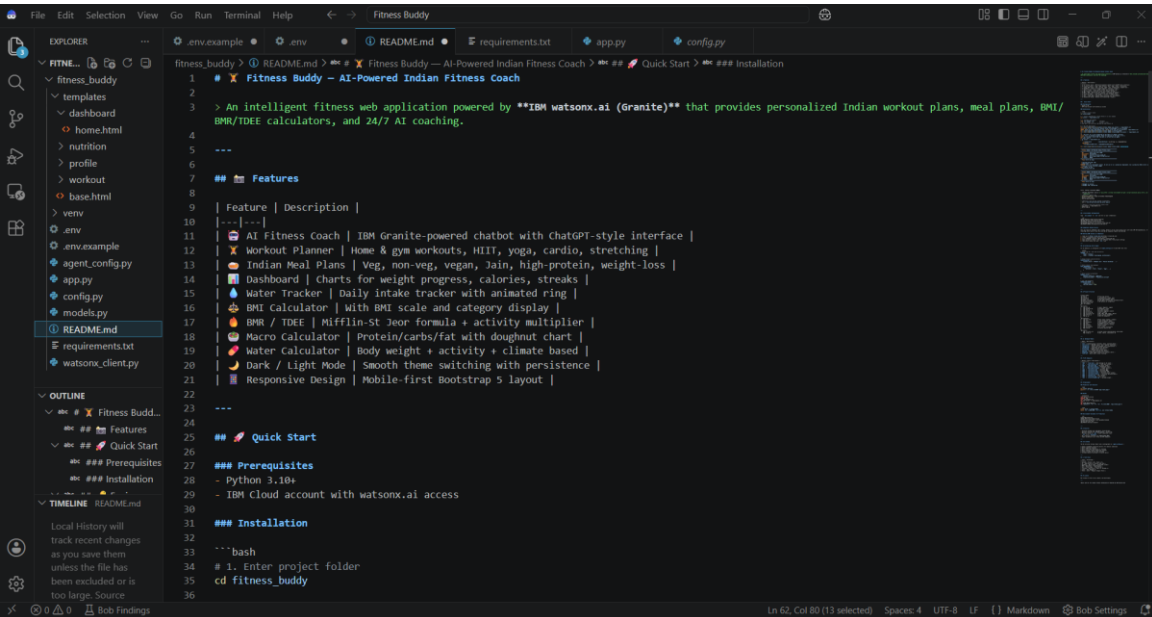
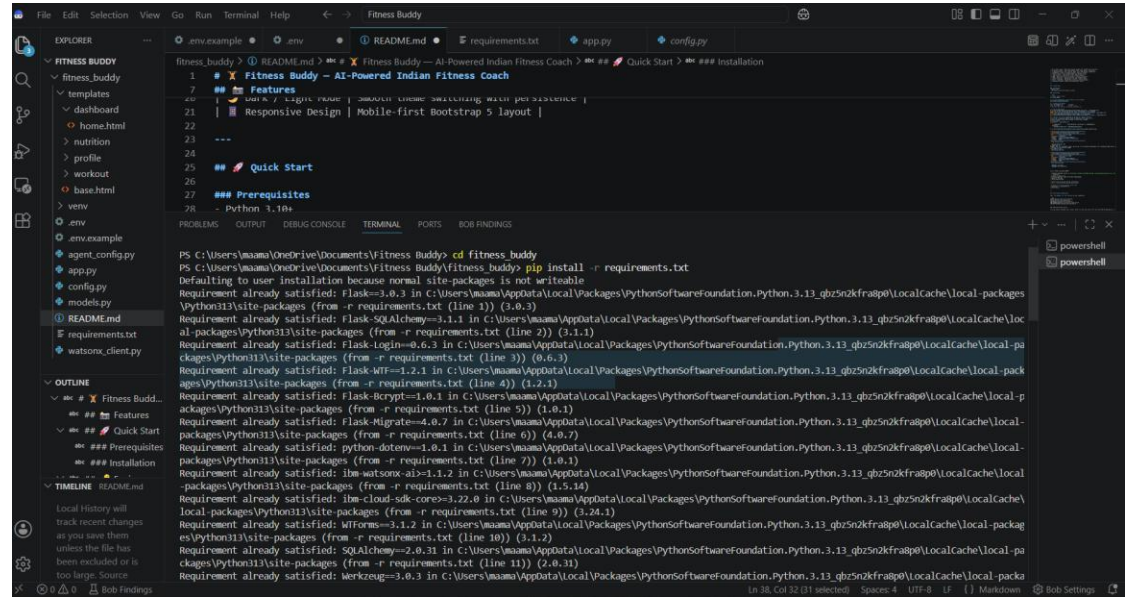
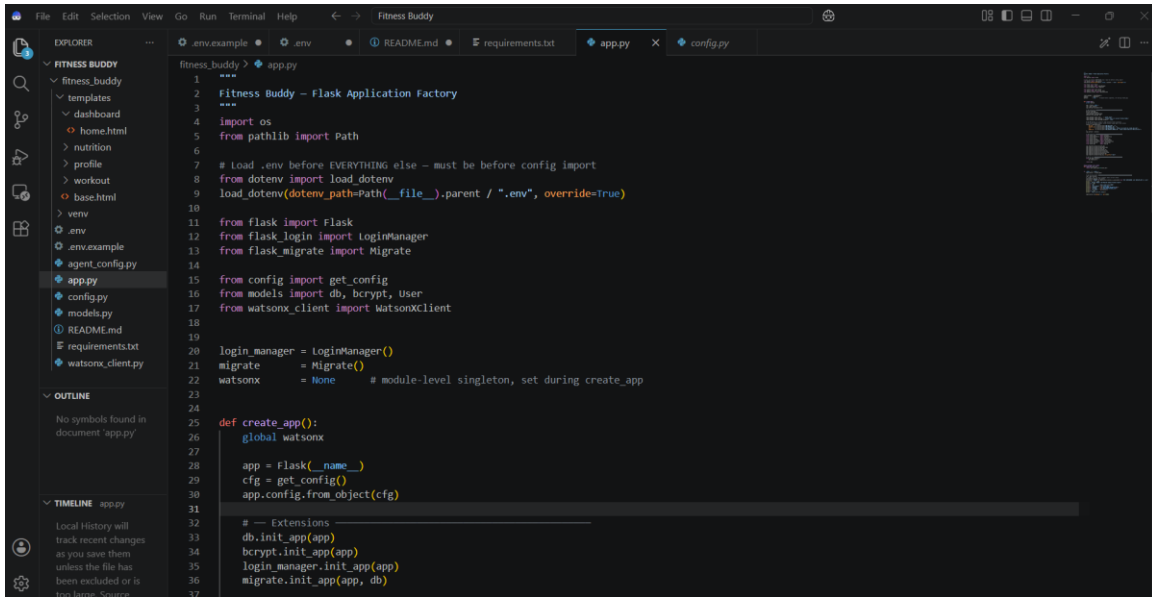
Today's Meals

BREAKFAST
2 whole eggs, 2 egg whites, 2 slices of whole wheat bread, and a glass of milk
(400 calories, 30g protein)
400 kcal · P 30.0g · C 45.0g · F 8.0g

LUNCH
hicken breast (120g), brown rice (100g), mixed vegetables (100g), and a bowl of dal
(500 calories, 40g protein)
500 kcal · P 40.0g · C 45.0g · F 8.0g

Indian Meal Plans

Other Applicable Screenshots:



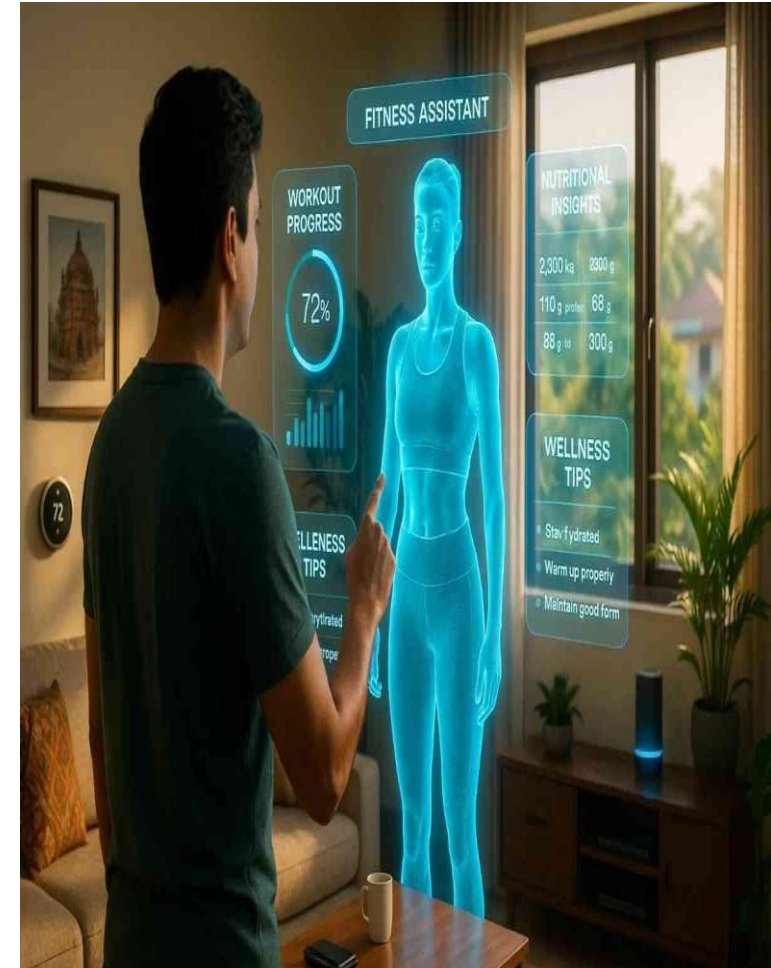
Role of Agentic AI in the solution

Role of Agentic AI in Fitness Buddy:

Agentic AI enables **Fitness Buddy** to function as an intelligent, autonomous fitness and wellness assistant rather than a simple chatbot. Built using **IBM watsonx.ai**, **IBM Orchestrate** and **IBM BOB**, it coordinates multiple specialized AI agents to deliver personalized health, nutrition, and workout recommendations.

Each agent performs a dedicated task—such as collecting user health information, generating customized workout plans, creating balanced Indian meal plans, calculating BMI and calorie needs, tracking daily progress, and answering fitness-related queries—while working collaboratively. This **multi-agent approach** ensures accurate, efficient, and goal-oriented guidance.

Agentic AI provides **context awareness** by understanding user details such as age, height, weight, fitness goals, activity level, dietary preferences, medical conditions, and workout experience. It continuously adapts recommendations based on user progress, daily activity, and feedback, enabling **real-time personalization**.



Novelty and Uniqueness:

The proposed Fitness Buddy stands out by combining **Agentic AI, and personalization** into a single intelligent system.

- 1.Multi-Agent Intelligence** – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.
- 2.Hyper-Personalization** – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.
- 3.Real-Time Feedback Loop** – Continuously adapts recommendations based on daily food logs and user behavior.
- 4.Multi-Modal Input** – Supports text, voice, and image-based meal tracking for better usability.
- 5.Preventive Healthcare Focus** – Goes beyond diet planning to provide disease-specific nutrition advice.
- 6.Scalable & Enterprise-Ready** – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Future Scope :

Future Scope of Fitness Buddy AI Agent

- 1.Integrate with wearable devices (smartwatches, fitness bands, and smart scales) to monitor heart rate, steps, calories burned, sleep, and activity in real time.
- 2.Use AI-powered computer vision to analyze exercise posture and provide real-time form correction and injury prevention.
- 3.Support multiple Indian languages such as Hindi, Punjabi, Tamil, Telugu, Marathi, Bengali, and Gujarati for wider accessibility.
- 4.Introduce an AI voice coach that guides users through workouts and provides live motivation and feedback.
- 5.Connect with food delivery and grocery platforms to recommend healthy Indian meal options and shopping lists.Provide adaptive workout and diet plans that automatically adjust based on user progress, performance, and health metrics.

Git Hub Link:

Please create public github repository (Enable Add README button while creating repository) with format Please test and Paste the working

Github URL – <https://github.com/yogeshchauhan1830/Fitness-Buddy.git>

Make sure you have Uploaded following three files into your github repository.

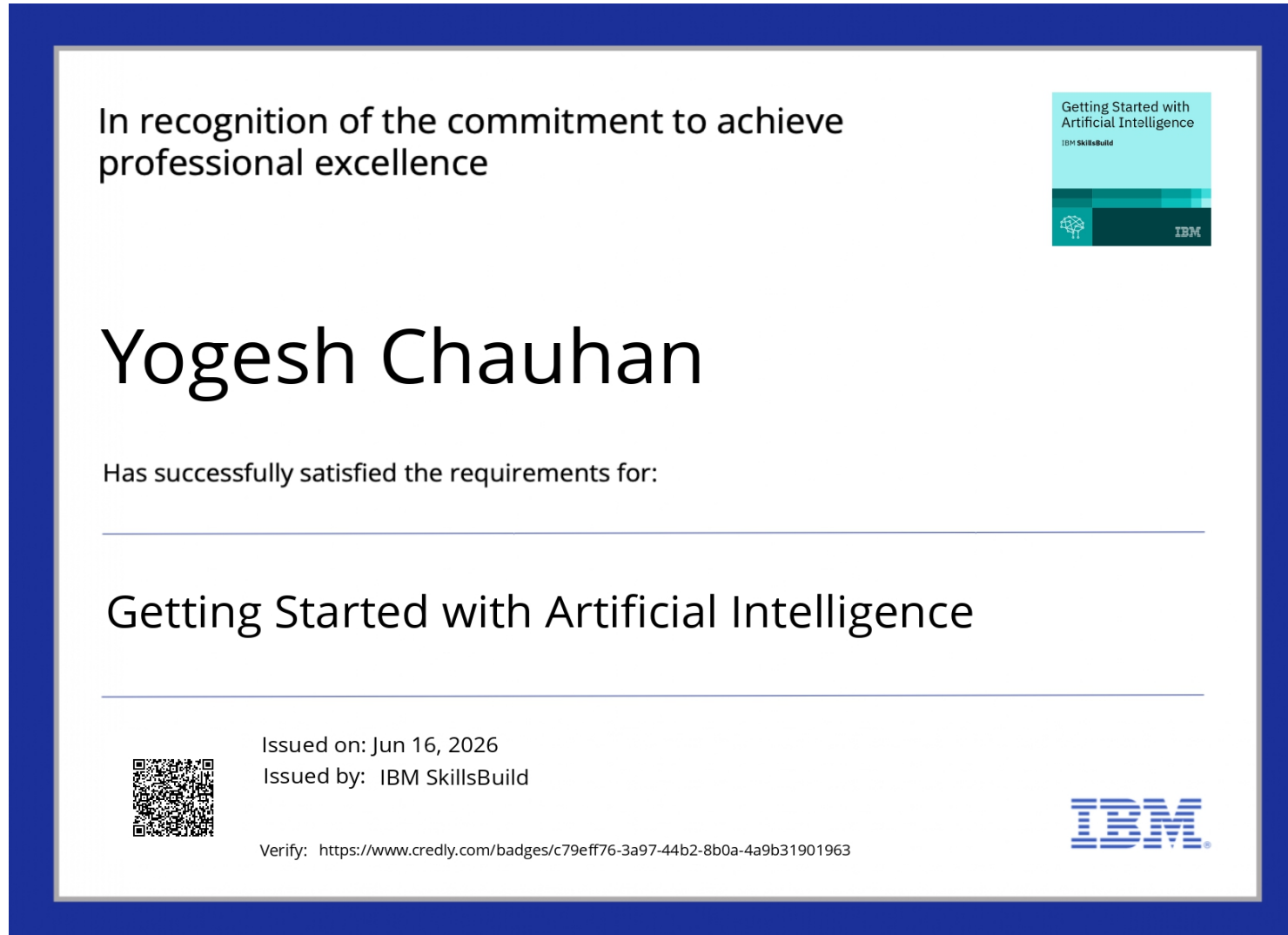
- 1) app.json file ,
- 2) yourproblemstatement.pdf file ,
- 3) your projectpresntation.pptx file ,
- 4)any other files.

Demo Video link-

https://drive.google.com/file/d/1EVcy8lHELV03_wTyv4ekR1VS75P11jKG/view?usp=sharing

Certificates Earned:

Please add your credly certificate-



Certificates Earned:

Add your IBM BOB Certificate

IBM SkillsBuild	Completion Certificate
	This certificate is presented to Yogesh Chauhan for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 17 Jun 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.